

11.2.

Známe-li matici D ze singulárního rozkladu A ,

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

pak uvažujeme matici

$$E = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 \end{pmatrix}.$$

Platí

$$D \cdot E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$E \cdot D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

2 předchozí úlohy:

$$U = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$V = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$

a tedy $EDE = E \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = E$

$$DED = D \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = D.$$

Uvažujeme nyní SVD rozklad matice A

$$A = UDV^T.$$

Ma-li platit

$$ABA = A$$

$$\text{a } BAB = B,$$

rozepíšme A podle rozkladu:

$$ABA = UDV^T B UDV^T$$

Zvolíme-li $B := VEU^T$, pak máme

$$\underbrace{UDV^T}_{I_3} \underbrace{VEU^T}_{I_4} UDV^T = UDEDV^T$$

Z předchozí úvahy $DED = D$, takže

$$= UDV^T = A.$$

Podobně

$$BAB = \underbrace{VEU^T}_{I_4} \underbrace{UDV^T}_{I_3} \underbrace{VEU^T}_E = VEDEU^T = VEU^T = B.$$

Zkontrolujme ještě, že AB a BA jsou symetrické

$$AB = UDV^T VEU^T \stackrel{V^T V = I_4}{=} UDEU^T \\ = U \cdot I_3 \cdot U^T = UU^T = I_3 \\ \uparrow DE = I_3$$

$$BA = VEU^T UDV^T = VEDV^T = V \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cdot V^T \\ \uparrow U^T U = I_3$$

$$= \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & 0 \\ 0 & -\frac{2}{\sqrt{6}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{pmatrix}$$

Matice B tedy splňuje všechny požadavky, pišme tedy

$$B = VEU^T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} \\ = \frac{1}{9} \begin{pmatrix} 3 & 3 & 3 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$